Cellular Automata in Materials Science Applications

2025 MITE Summer Program
Hosted by the Wang Materials Group

The University of Texas at Austin
McKetta Department of Chemical Engineering
Cockrell School of Engineering

TEXAS MATERIALS INSTITUTE
Materials Science & Engineering

Today’s plan

What is Materials Science and Engineering?

- Materials Science Tetrahedron
- Examples: Spinodal Decomposition

What is computation and simulation?

What is cellular automata?

- Conway’s Game of Life
- Spinodal decomposition

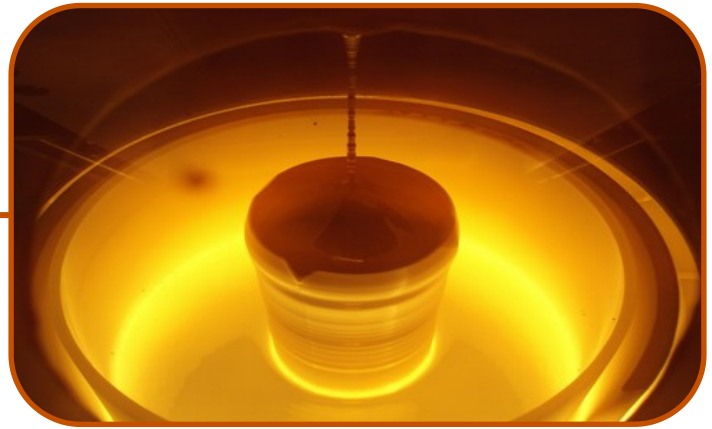
What is materials science and engineering?

Discovering, understanding, and designing materials (often times solids) through the study and application of processing-structure-property-performance relationships

Processing → Structure → Properties → Performance

What is materials science and engineering?

How is the material actually made in a lab/factory or adapted from nature?

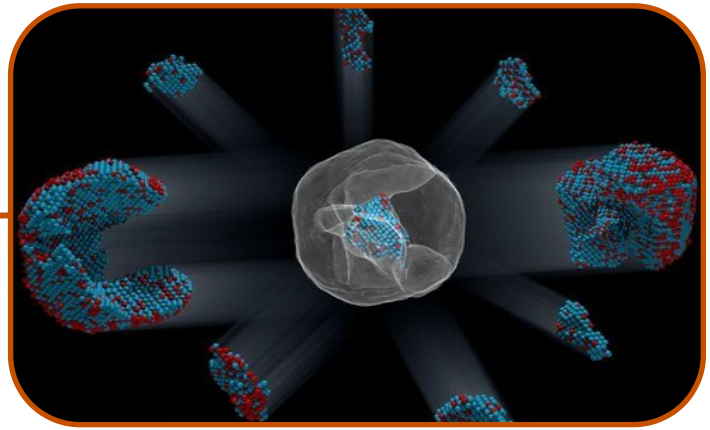


Processing → Structure → Properties → Performance

Adapted from Modern Materials Technologies, Image from Omnidyne

What is materials science and engineering?

How are the internal components arranged?



Processing → Structure → Properties → Performance

Adapted from Modern Materials Technologies, Image from Chong, L. “Bringing atomic mapping to the mainstream” Phys Org (2017)

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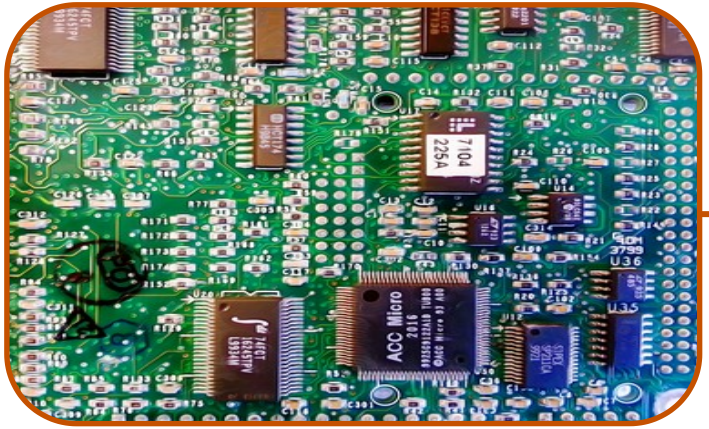

How does it respond to an external stimulus?

- Mechanical
- Electrical
- Thermal
- Magnetic
- Optical
- Chemical

Processing → Structure → Properties → Performance

Adapted from Modern Materials TechnologiesModern Materials Technologies

What is materials science and engineering?

How does the material help the user/world?

Processing → Structure → Properties → Performance

Image from Alset International

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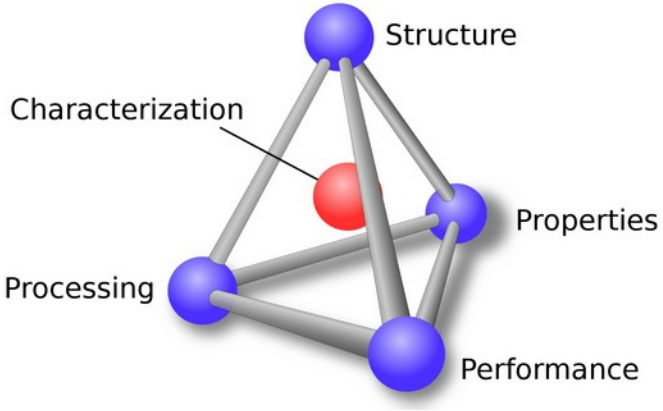
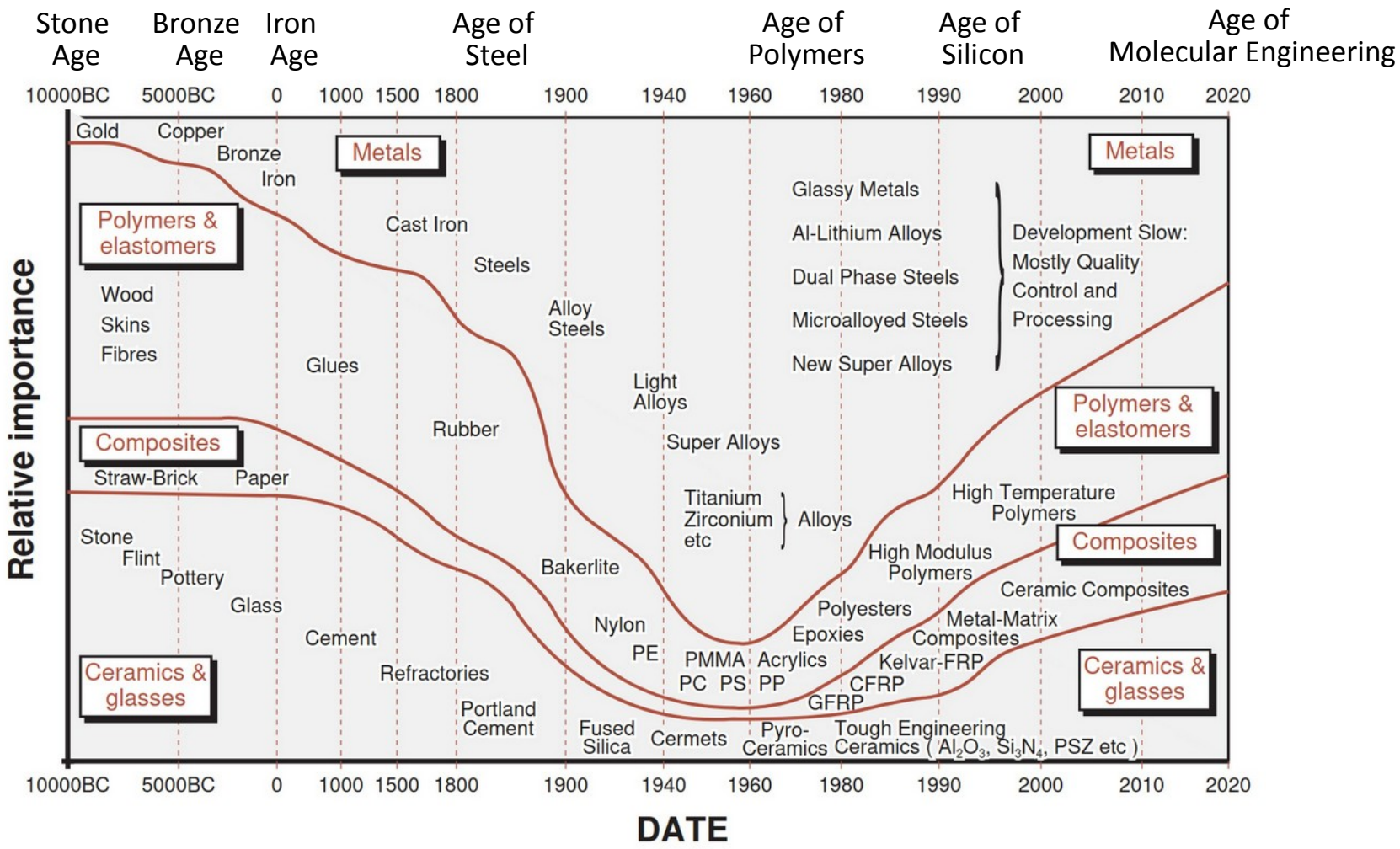
Processing → Structure → Properties → Performance



Why study materials? (Think-pair-share)

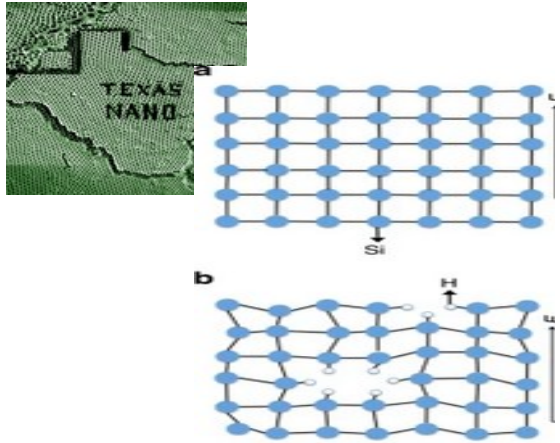


Adapted from Ashby M.F., 2005. Materials Selection in Mechanical Design, 3rd ed

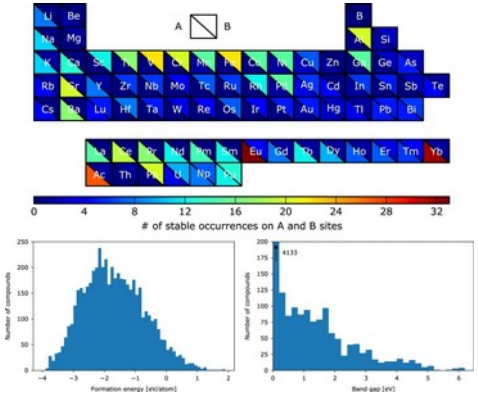
Examples of Materials Science and/or Engineering

3D printing soft bio-polymers
Nelson Lab, University of Washington

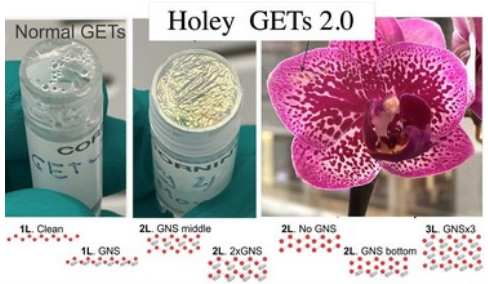
Using viruses as templates to make more efficient solar cells
Belcher Lab, MIT



Using defects in nanoparticles for high-efficiency, low-loss nanophotonics
Korgel & Zheng Research Groups, UT Austin

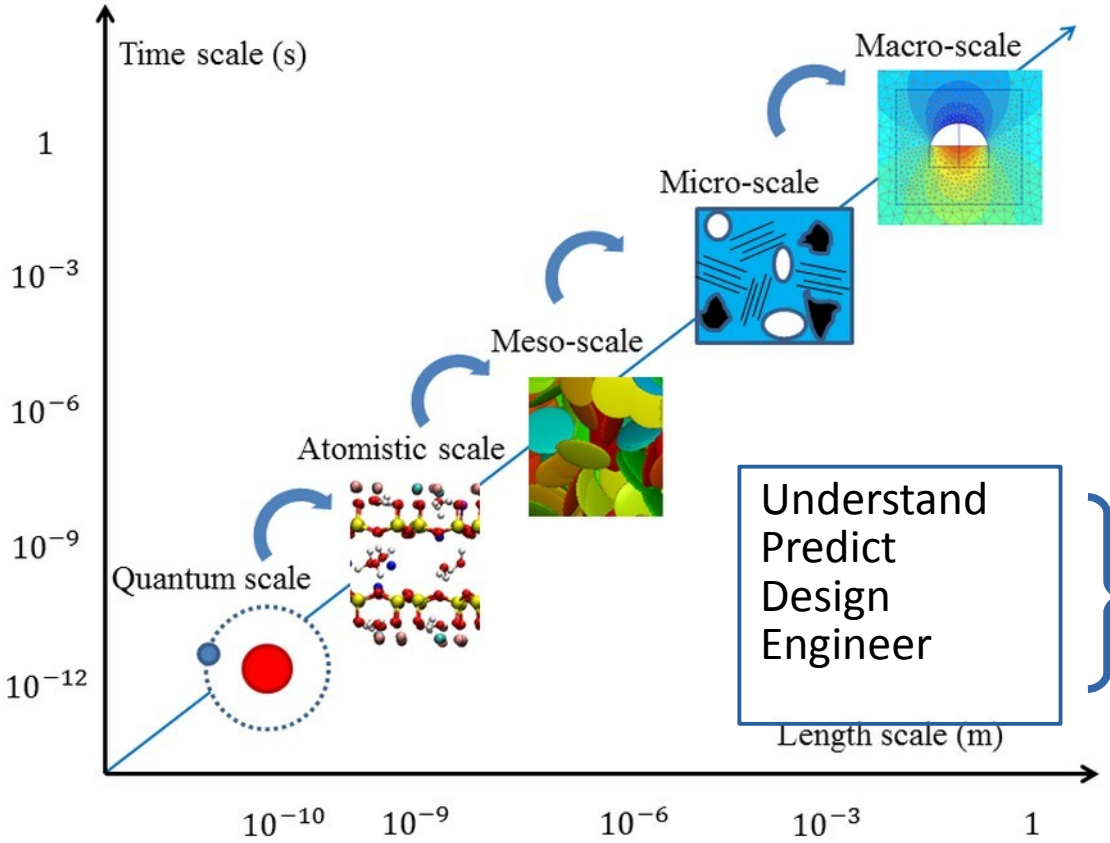


Using supercomputers to discover new and ideal oxides
Wolverton Research Group, Northwestern



Breathable graphene tatoos for personalized healthcare
Akinwande Nano Research Group, UT Austin

What is Modeling and Simulation in Materials Science?



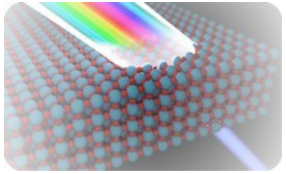
Understand
Predict
Design
Engineer

Material properties
New materials

TACC
TEXAS ADVANCED COMPUTING CENTER

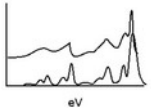
Research Overview and Philosophy

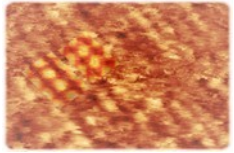
First-principles calculations



Enabled by HPC resources

Experimental collaboration

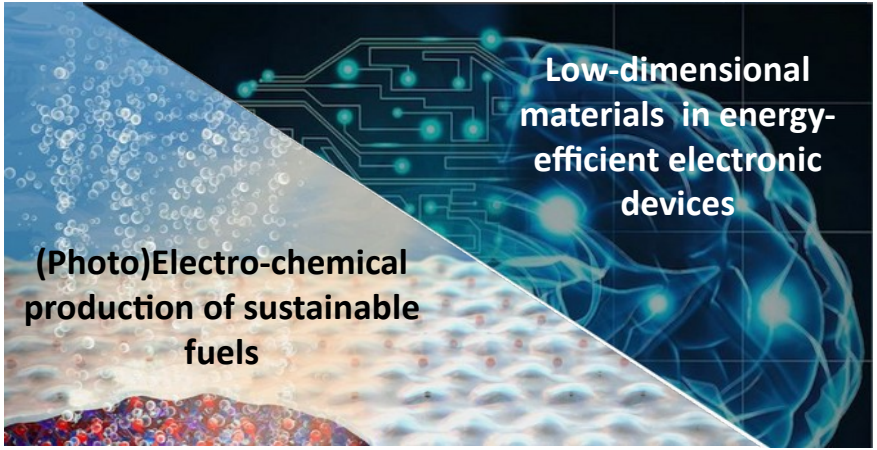




Spectroscopies, microscopies, ...

Wang Modern Materials

Harnessing Materials Imperfections for Energy Sustainability



Conway's Game of Life

Variation of cellular automata

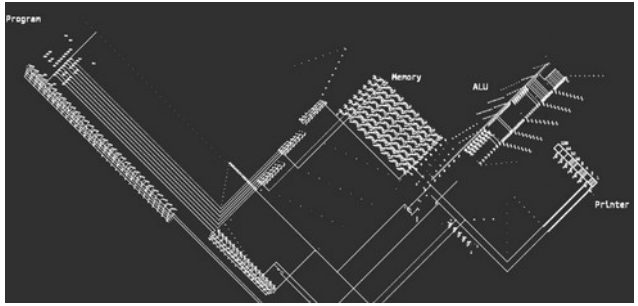
Invented by Cambridge mathematician John Conway (~1969)

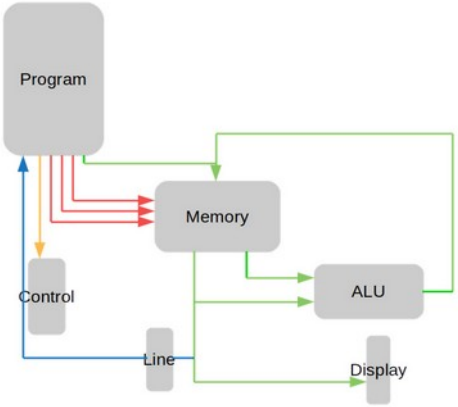
"He spent most of his time playing games, inventing games and reinventing rules to games he found boring, such as chess."

The Guardian (July 15, 2023)

"no-player never-ending" game

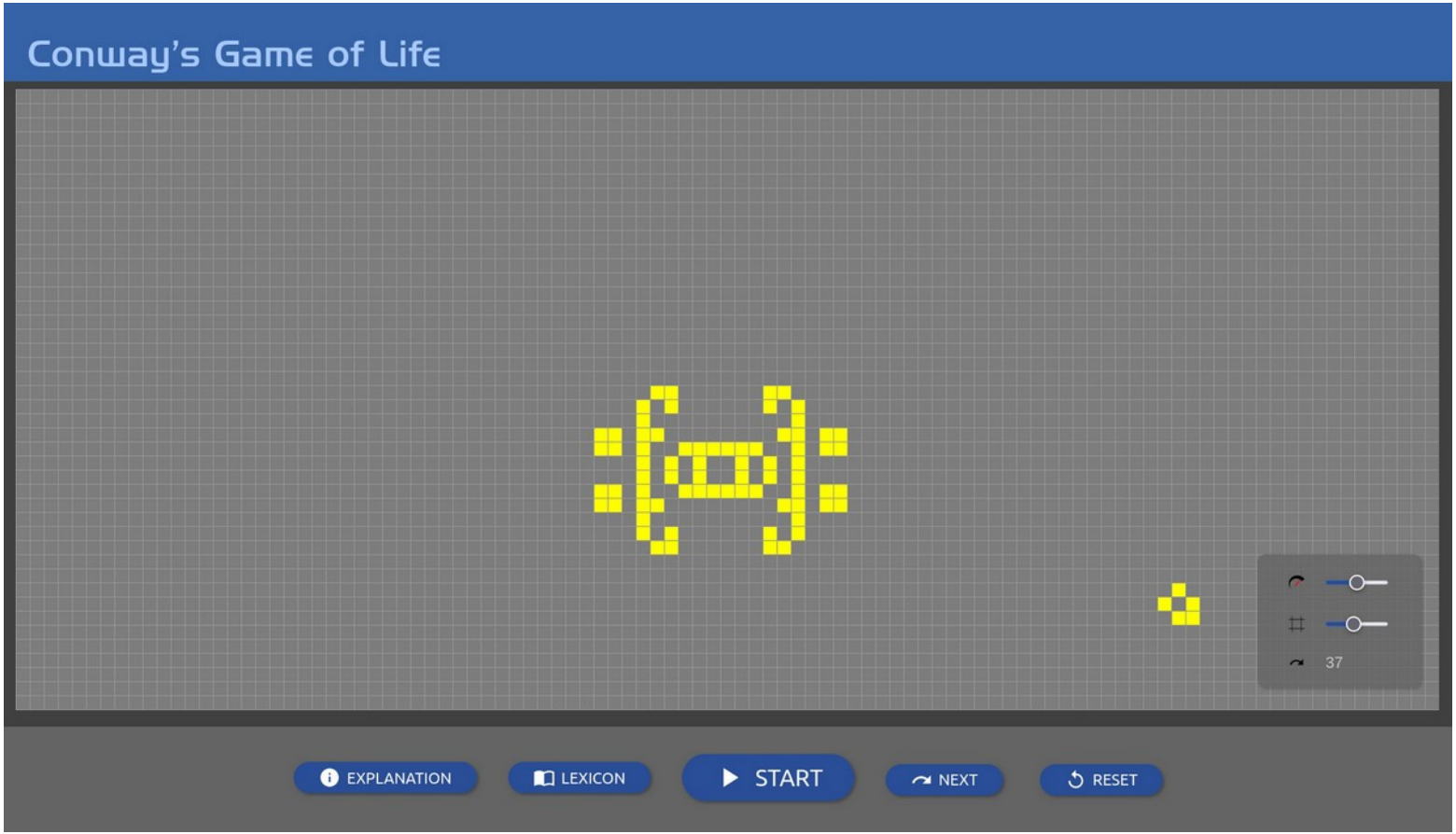
Known to be Turing-complete





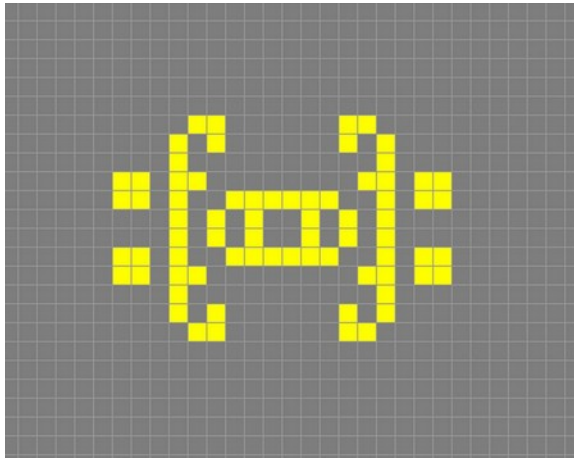
https://www.nicolasloizeau.com/gol-computer

Conway's Game of Life



https://playgameoflife.com/

Conway's Game of Life



On a 2D square grid, at *t*=0, some cells are 'alive' (yellow) and the rest are 'dead' (grey).

At each time step, the value of the cell is updated:

- Alive cells stay alive if they have **two** or **three** living neighbors.
- Dead cells are revived to alive cells if there are **exactly three** living neighbors (imitating reproduction).
- Alive cells die if there are **fewer than two** living neighbors (underpopulation) or **more than three** living neighbors (overpopulation).

A more realistic example: Spinodal Decomposition

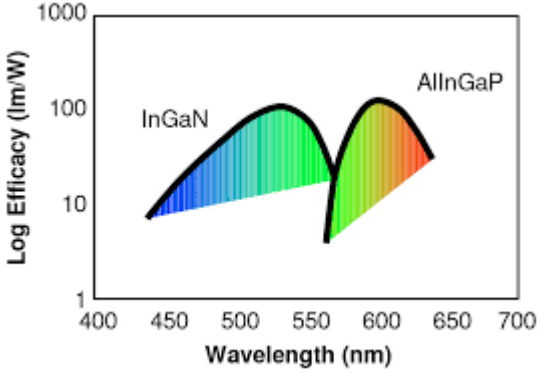
Alloys

Brass (Cu, Zn, ...)

Bronze (Cu, Sn,...)

Steel (Fe, C,...)

InGaN LEDs



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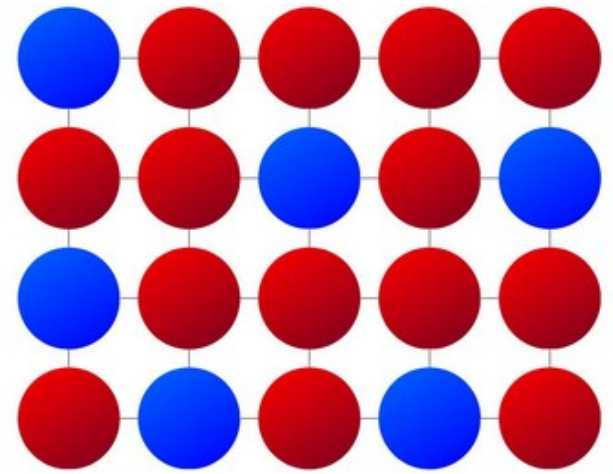
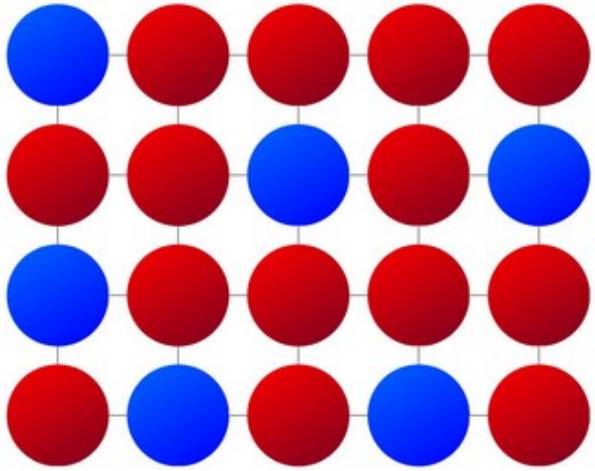
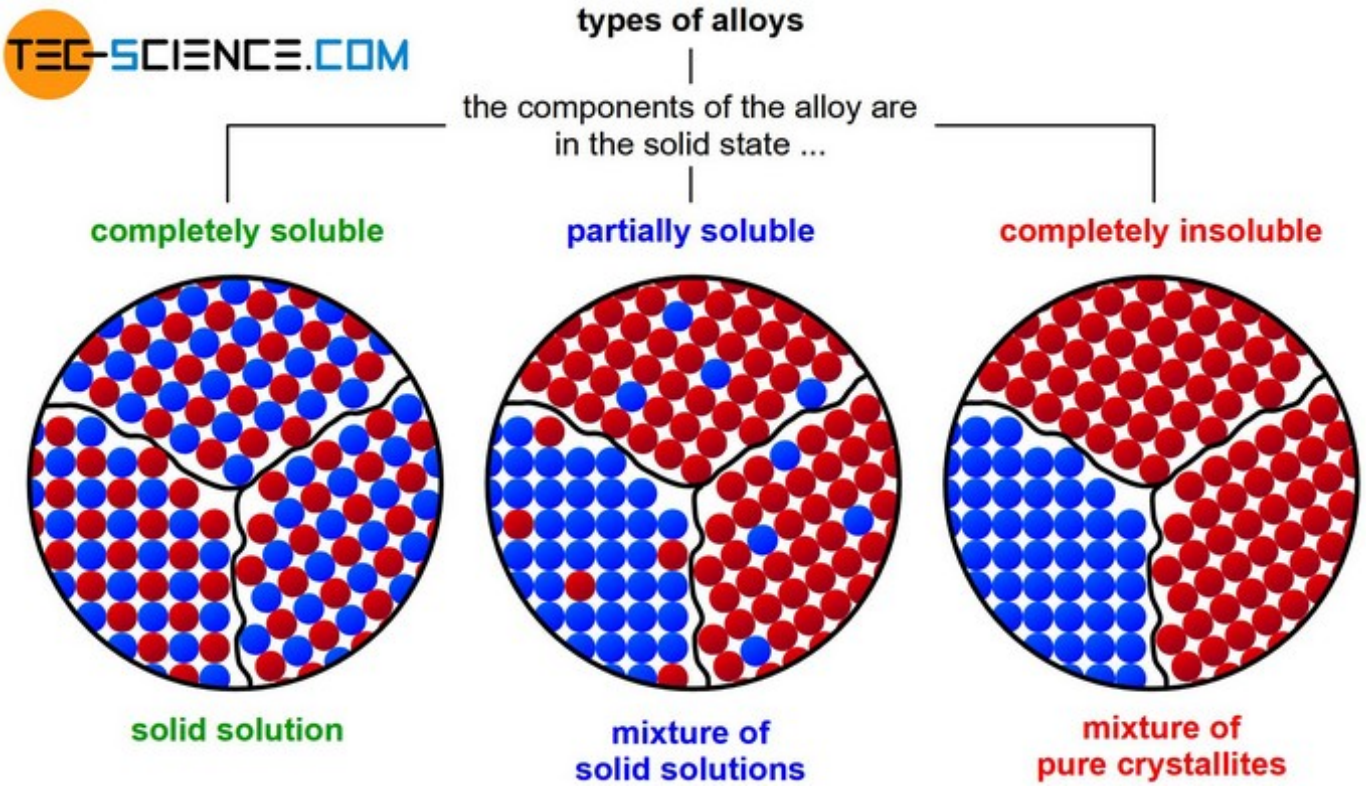


https://www.tec-science.com/material-science/alloys/typs-of-alloys/

Many alloys are not perfect solid solutions





Mixture of “phases”

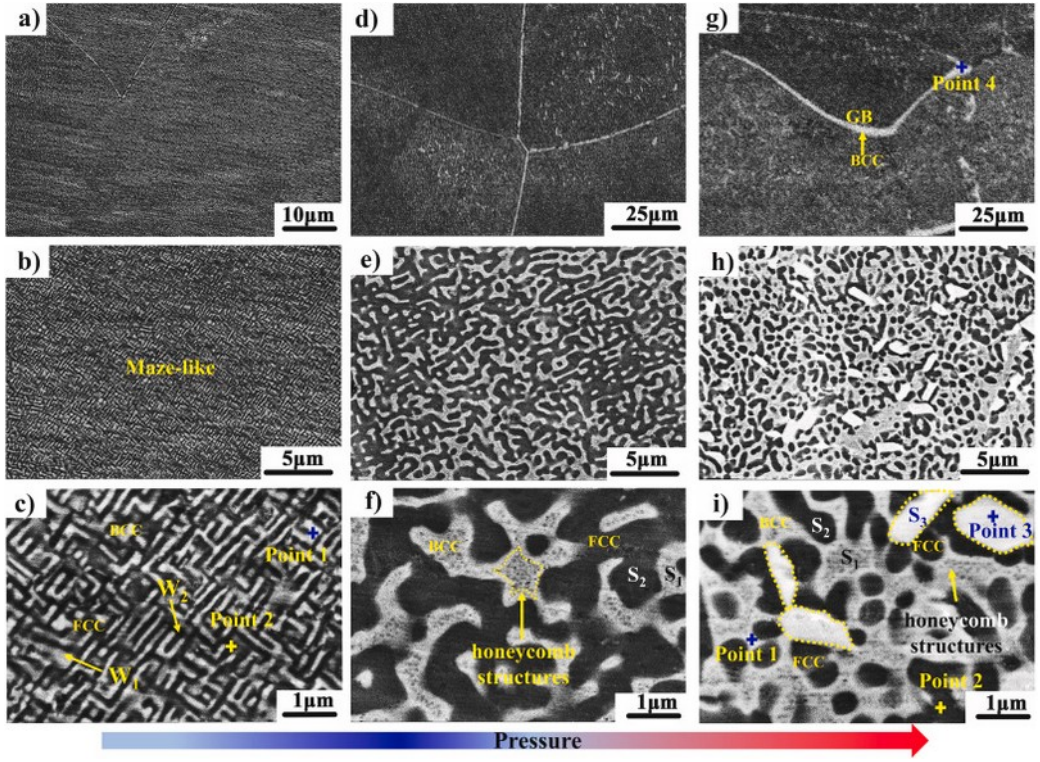
https://www.tec-science.com/material-science/alloys/typs-of-alloys/

Spinodal Decomposition

Spinodal Decomposition

is spontaneous phase separation below a critical temperature,
often limited by diffusion

Microstructure evolution with pressure for CoCrFeNiAl alloy



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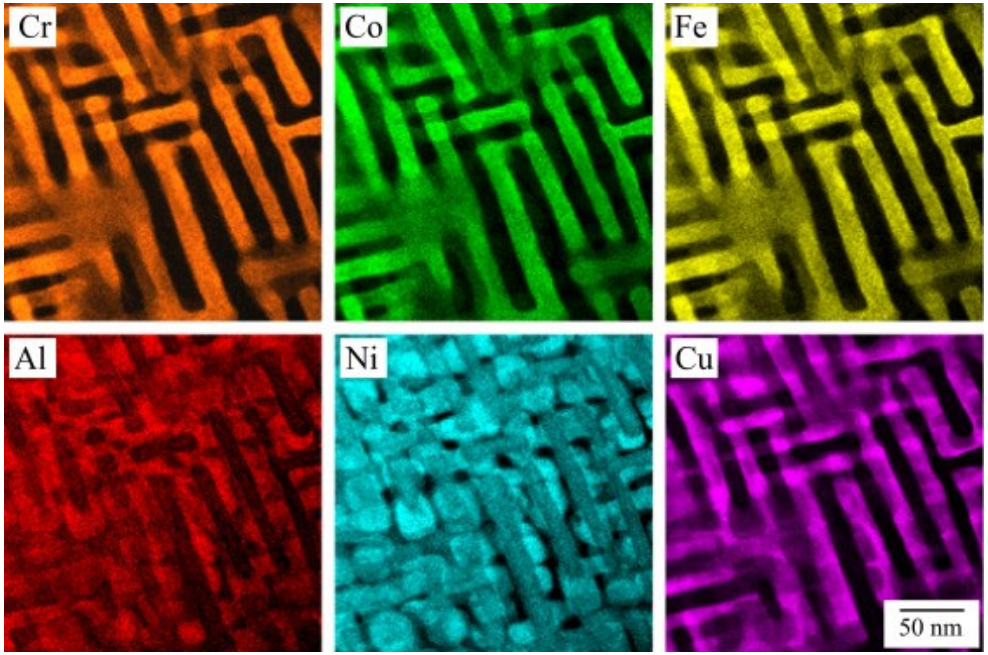

STEM-EDX image of spinodal decomposition of high-entropy alloy Al0.5CrFeCoNiCu

A brief intro to Python

Python is a high-level, general-purpose programming language.

Its design philosophy emphasizes code readability.

This makes Python a great language to learn basic programming concepts, which can be used to learn other programming languages.

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Just like any other language, there are rules on how to write code.
We call these rules the ***syntax***.

For example, consider the role of the comma in these two sentences:

Let's eat grandma vs Let's eat, grandma

Computers need *exact* instructions, so poor syntax (e.g., typos) will lead to code errors,

Let’s look at a Python code

```python
from dataclasses import dataclass
from pathlib import Path

# for compatibility across different versions of Python
try:
    # compatible with Python >= 3.11
    import tomllib
except ImportError:
    import tomli as tomllib

# import the list of possible patterns
PATTERNS_FILE = Path(__file__).parent / "patterns.toml"

# The decorator @dataclass automatically generates special methods
# like __init__ and __repr__ to user-defined classes, which saves us effort!
# see documentation: https://docs.python.org/3/library/dataclasses.html
@dataclass
class Pattern:
    """ Pattern class in Conway's Game of Life

        A square array of 'dead' and 'alive' cells is tracked at each iteration.
        In this implementation, the array is initiated with 'dead' cells and
        we keep track of coordinate index of the 'alive' cells only as a list of tuples

    """

    name: str
    alive_cells: set[tuple[int, int]]
```

To make code easier to read for humans, we often put in descriptive ***comments***

Two types of comments: line comments (marked with #) and block comments (marked with “”””)

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Python is modular.
We can import
different ***packages***

Let’s try a little coding and simulating!